EAMHP

Excel-Based Avionics Mechanical & Health Prediction

TECHNICAL REQUIREMENTS DOCUMENT

TRD — Version 1.1.0

Document ID	EAMHP-TRD-001
Version	1.1.0
Date	March 2026
Author	Nikhilesh (Nikhilesh Avionics Projects)
Status	UNRELEASED
Classification	
Aircraft Reference	Airbus A320 Family (CFM56-5B / IAE V2500-A5)
Platform	Microsoft Excel with VBA / Macros (.xlsm)

1. Introduction

EAMHP (Excel-Based Avionics Maintenance & Health Prediction) is a desktop-grade avionics maintenance tool built on Microsoft Excel with VBA/Macro automation. It is designed to assist Aviation Maintenance Engineers (AMEs) and avionics trainees in identifying, diagnosing, and actioning engine health faults on the Airbus A320 family powered by CFM56-5B and IAE V2500-A5 engines.

This Technical Requirements Document (TRD) defines the functional architecture, data structures, system logic, interface specifications, and technical constraints that govern the EAMHP V1.1.0 implementation.

1.1 Scope

This document covers:

- Fault detection logic using multi-parameter engine performance data
- Reference data architecture for CFM56-5B and IAE V2500-A5 engine variants
- Fault matrix design covering 2-parameter and 3-parameter combinations
- Corrective and preventive action mapping per fault category
- Engine-specific AMM/TSM procedure linkage for CFM56-5B and V2500
- Excel workbook sheet architecture and data-flow logic
- Input validation, data integrity constraints, and error handling

1.2 Definitions and Abbreviations

Abbreviation	Full Form
AME	Aviation Maintenance Engineer
AMM	Aircraft Maintenance Manual
TSM	Trouble Shooting Manual
EAMHP	Excel-Based Avionics Maintenance & Health Prediction
EGT	Exhaust Gas Temperature
EPP	Engine & Performance Parameters
FADEC	Full Authority Digital Engine Control
FOD	Foreign Object Damage
HMU	Hydro-Mechanical Unit
HPC	High Pressure Compressor
LPC	Low Pressure Compressor
MRO	Maintenance, Repair and Overhaul
N1	Fan / Low Pressure Compressor Rotational Speed (%)
N2	High Pressure Compressor Rotational Speed (%)

Abbreviation	Full Form
TOGA	Take-Off / Go-Around
VBA	Visual Basic for Applications
xlsm	Excel Macro-Enabled Workbook

2. System Architecture

EAMHP operates as a single-file Excel workbook (.xlsm) containing six functional sheets integrated through VBA macros and Excel formula logic. The architecture follows a layered design pattern: input collection, reference lookup, fault evaluation, and output generation.

2.1 Workbook Sheet Architecture

Sheet Name	Sheet Type	Function Description
InputData	Operational	Accepts timestamped engine parameter readings (Parameter, Unit, Value, Engine Type). Automatically evaluates Fault Status against reference tolerances via formula lookup.
ReferenceData	Static Reference	Stores all Engine Performance Parameter (EPP) definitions including lower/upper tolerance bounds and units for CFM56-5B and IAE V2500-A5.
FaultMatrixEngine	Logic Reference	Contains the fault identification matrix mapping fault IDs to parameter pairs with state templates (HIGH/LOW deviations).
DataValidation	Validation Reference	Provides dropdown validation lists for Parameter Names, Unit types, and Engine Variants used in InputData.
ProjectGuide	Documentation	Internal reference sheet listing abbreviations and project glossary for AME users.
Sheet2	Reserved	Placeholder sheet reserved for future expansion (V1.0.1+).

2.2 Data Flow Architecture

The system processes data in the following sequential flow:

1. The user enters engine parameter data (Timestamp, Engine Type, Parameter, Unit, Value) into the InputData sheet.
2. Excel VLOOKUP/INDEX-MATCH formulas query ReferenceData to retrieve Upper and Lower Tolerance for the entered Parameter + Engine Type combination.
3. The Fault Status column evaluates whether the entered Value falls outside tolerance bounds and returns fault flag (FAULT / NORMAL).

4. Faulted parameters are cross-referenced against FaultMatrixEngine to identify applicable Fault_ID combinations.
5. Output includes Fault Category, Corrective Action, Preventive Action, and engine-specific AMM/TSM procedures for both CFM56-5B and V2500 variants.

3. Engine Performance Parameter Reference Data

The ReferenceData sheet stores 16 engine performance parameters (EPP1–EPP16+) covering both CFM56-5B and IAE V2500-A5 variants. All tolerance values are based on standard Airbus A320 family AMM references under ISA Standard Atmosphere / Sea Level conditions.

3.1 Monitored Parameters

Parameter ID	Parameter Name	Engine Variant	Unit	Lower Tolerance	Upper Tolerance
EPP1	N1 Idle	CFM56-5B	%	18.1	24.2
EPP2	N1 Idle	IAE V2500-A5	%	19.0	21.0
EPP3	N1 Max	CFM56-5B	%	98.8	109.2
EPP4	N1 Max	IAE V2500-A5	%	98.8	109.2
EPP5	N2 Idle	CFM56-5B	%	55.1	65.1
EPP6	N2 Idle	IAE V2500-A5	%	57.0	63.0
EPP7	N2 Max	CFM56-5B	%	99.8	110.3
EPP8	N2 Max	IAE V2500-A5	%	99.8	110.3
EPP9	EGT Idle	CFM56-5B	°C	370.5	456.8
EPP10	EGT Idle	IAE V2500-A5	°C	380.0	472.5
EPP11	EGT Max (TO/GA)	CFM56-5B	°C	603.3	666.8
EPP12	EGT Max (TO/GA)	IAE V2500-A5	°C	617.5	682.5
EPP13	EGT Max Continuous	CFM56-5B	°C	579.5	640.5
EPP14	EGT Max Continuous	IAE V2500-A5	°C	579.5	640.5
EPP15	Oil Pressure Min	CFM56-5B / V2500	psi	13.0	75.0
EPP16	Oil Quantity Min	CFM56-5B / V2500	qt/L	6.5	22.0

3.2 Supported Unit Types

The DataValidation sheet supports the following engineering unit types for parameter entry:

- Percentage: % (N1, N2 speed parameters)
- Temperature: °C, °F (EGT parameters)
- Pressure: psi, bar, kPa, kg/cm², mmHg, inHg
- Flow: kg/h, lb/h, cfm
- Speed: RPM, kts

- Distance/Altitude: ft, m, mm
- Electrical: Amps, Volts, Hz, Ohms
- Torque: Nm, kg/cm
- Volume: gallons, liters
- Time: seconds, minutes, hours

4. Fault Matrix Design

EAMHP implements a comprehensive fault identification matrix that correlates parameter deviations to fault signatures. The matrix covers 16 engine performance parameters yielding 2-parameter and 3-parameter fault combination sets.

4.1 Fault Combination Statistics

Parameter Set	Total Fault IDs	Sheet Reference
2-Parameter Combinations	120 (F0001–F0120)	FaultMatrixEngine, EAMHP_fault_combinations_2_3_params
3-Parameter Combinations	560 (F0121–F0680)	EAMHP_fault_combinations_2_3_params
Total Fault Combinations	680	All sheets
Fault Categories	5	Mixed Report sheet
Engine Variants Covered	2 (CFM56-5B, IAE V2500-A5)	FaultMatrixEngine

4.2 Fault Categories

Fault Category	Description	Key Parameters
General Engine Performance Anomaly	Deviations in core spool speeds and thrust output indicating FADEC/sensor faults.	N1, N2, Max Thrust, Starter Limit
General Engine Overtemperature	EGT exceedances requiring hot section inspection and fuel schedule verification.	EGT Idle, EGT Max TOGA, EGT Max Continuous
Oil System Malfunction	Oil pressure/temperature/quantity deviations indicating pump, filter, or leak faults.	Oil Pressure Min/Max, Oil Temp, Oil Quantity
Compressor Efficiency Loss	Fuel flow and speed anomalies indicating compressor blade/wash/FOD issues.	N1 Idle/Max, Fuel Flow Cruise
Combustion Inefficiency	Combined fuel, EGT, and performance deviations indicating combustor issues.	EGT, Fuel Flow, N1/N2

4.3 Fault ID Format

All fault IDs follow the format: F[4-digit sequential number], e.g., F0001, F0121, F0680. Fault IDs are unique across the entire fault matrix and are referenced consistently across all supporting sheets.

5. Corrective and Preventive Action Mapping

Each fault ID is mapped to a Fault Category, a Corrective Action, and a Preventive Action within the EAMHP_fault_matrix_with_actions.xlsx reference dataset. This mapping forms the maintenance decision support core of EAMHP.

5.1 Action Mapping Schema

Field	Description	Example
Fault_ID	Unique fault identifier	F0004
Parameters_Involved	Comma-separated triggering parameters	N1 Idle, EGT Idle
Fault_Category	Classified fault type	General engine overtemperature
Corrective_Action	Immediate maintenance action required	Reduce thrust, inspect hot section, borescope
Preventive_Action	Scheduled maintenance to prevent recurrence	Monitor EGT trend, schedule hot section checks

5.2 Sample Action Mapping

Fault_ID	Category	Corrective Action
F0004	Overtemperature	Reduce thrust, inspect hot section, verify fuel schedule, borescope inspection
F0007	Oil System	Inspect oil pump, check oil filter & cooler, inspect for leaks, verify oil quantity
F0013	Compressor	Inspect compressor blades, check IGV & bleed valves, perform compressor wash, FOD check
F0001	Performance	Perform engine performance check, inspect sensors, verify FADEC and control systems

6. Engine-Specific AMM/TSM Procedure Linkage

EAMHP V1.1.0 integrates engine-specific maintenance procedures sourced from the Aircraft Maintenance Manual (AMM) and Trouble Shooting Manual (TSM) for both engine variants. This data is stored in the EAMHP_fault_matrix_with_engine_procedures.xlsx reference file organized by fault category.

6.1 Procedure Coverage

Fault Category Sheet	Row Count	Procedure Coverage
Mixed Report (All Categories)	680 records	Full fault matrix with both engine procedures
Oil_System_Malfunction	497 records	CFM56-5B and V2500 oil system specific
Engine_Performance	45 records	FADEC, sensor, and performance troubleshooting
Engine_Temp_Over_temperature	85 records	Hot section, borescope, and EGT procedures
Compressor_Efficiency_Loss	11 records	Compressor wash, IGV, FOD, VSV procedures
Combustion_inefficiency	42 records	Combustor, fuel nozzle, and spray dome procedures

6.2 Procedure Format

Engine-specific procedures are stored in two columns per fault record:

- CFM56_5B_Procedure: AMM/TSM action referencing CFM56-5B specific components (HMU, LPC/HPC, scavenge system, etc.)
- V2500_Procedure: Equivalent action referencing V2500-A5 specific components (FADEC, lubrication pump, combustor dome, etc.)

Example: For F0007 (Oil System Malfunction):

CFM56-5B: Inspect oil pump, oil filter, and oil cooler; check scavenge system and leaks.

V2500: Inspect lubrication pump, oil filter bypass valve, verify oil cooler performance and check for leaks.

7. Input Data Specification

The InputData sheet accepts the following fields for each engine parameter entry. All fields except Timestamp are mandatory.

Field Name	Data Type	Required	Validation Source	Description
TimeStamp	DateTime	Optional	Free entry	Date/time of parameter reading
Engine Type	Text (Dropdown)	Yes	DataValidation	CFM56-5B or IAE V2500-A5
Parameter	Text (Dropdown)	Yes	DataValidation (16 EPPs)	Engine parameter name
Unit	Text (Dropdown)	Yes	DataValidation (29 units)	Engineering unit of measurement
Value	Numeric	Yes	Manual entry	Measured parameter value

Field Name	Data Type	Required	Validation Source	Description
Upper Tolerance	Numeric (Auto)	Auto	ReferenceData VLOOKUP	Upper bound from reference
Lower Tolerance	Numeric (Auto)	Auto	ReferenceData VLOOKUP	Lower bound from reference
Fault Status	Text (Auto)	Auto	Formula evaluation	FAULT or NORMAL result

7.1 Fault Status Logic

The Fault Status column uses the following evaluation logic:

- If Value > Upper Tolerance: Status = FAULT (HIGH deviation)
- If Value < Lower Tolerance: Status = FAULT (LOW deviation)
- If Lower Tolerance <= Value <= Upper Tolerance: Status = NORMAL
- If lookup fails (no reference match): Status = #N/A (data entry error)

8. Technical Constraints

8.1 Platform Constraints

- EAMHP V1.1.0 is designed exclusively for Microsoft Excel 2016 or later (Windows)
- Macro execution requires VBA to be enabled in Excel Trust Center settings
- File format must be maintained as .xlsm (Excel Macro-Enabled Workbook)
- macOS Excel compatibility is not guaranteed due to VBA implementation differences
- Google Sheets is not supported as it does not support VBA macros

8.2 Data Constraints

- Maximum 16 engine performance parameters per engine variant in V1.1.0
- Fault matrix covers 2-parameter and 3-parameter combinations only (max 3 simultaneous fault parameters)
- Fault IDs are sequentially assigned and must not be duplicated across sheets
- ReferenceData lookup requires exact string match on Parameter Name and Engine Variant fields

8.3 Performance Constraints

- Workbook file size target: under 5 MB for V1.1.0
- Fault lookup response time: instantaneous (formula-driven, no VBA delay required)
- Maximum InputData rows supported: 1000 parameter entries per session

9. Version History

Version	Date	Author	Changes
1.1.0	TBD	Nikhilesh	Initial release of EAMHP TRD. Full architecture documentation for V1.1.0 covering 680 fault combinations, dual-engine variant support, AMM/TSM procedure linkage, and Excel workbook design.
1.2.0	TBD	Nikhilesh	Planned: Additional parameters, expanded 3-parameter combinations, A320neo engine variant support.
1.3.0	TBD	Nikhilesh	Planned: Boeing 737 MAX / 737 Classic engine family expansion.